Sihong Wu

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Education

- **Ph.D. in Geophysics** **2016 – 2021**
  Peking University *Beijing, China*
  Thesis: *De-noising and inversion of transient electromagnetic data based on the deep learning methods*
  Advisor: Dr. Qinghua Huang
- **B.S. in Geophysics** **2012 – 2016**
  University of Science and Technology of China *Anhui, China*
  Enrolled in Special Class for the Gifted Young (*for age 16 and under*)

Appointments

- **Postdoctoral Researcher** **2025 – Present**
  Massachusetts Institute of Technology *MA, USA*
  Supervisor: Dr. Laurent Demanet, Dr. Aimé Fournier
  Geothermal system characterization; Uncertainty quantification; Borehole acoustic logging.
- **Postdoctoral Researcher** **2023 – 2025**
  University of Houston *TX, USA*
  Supervisor: Dr. Jiajia Sun, Dr. Jiefu Chen
  Uncertainty quantification; Oceanic crust formation; Natural hydrogen exploration; Geothermal system imaging; Seawater intrusion monitoring.
- **Postdoctoral Researcher** **2021 – 2023**
  Peking University *Beijing, China*
  Supervisor: Dr. Qinghua Huang
  Machine learning in electromagnetic geophysics; Uncertainty quantification; Real-time subsurface imaging.

Research Interests

- **Artificial intelligence in geophysics:** *machine learning, Bayesian inference, and uncertainty quantification for large-scale electromagnetic, seismic, gravity, and magnetic data integration.*
- **Sustainable resources and energy systems:** *exploration and monitoring of geothermal, critical mineral, and natural hydrogen and geological carbon storage systems*
- **Hydrogeophysics and near-surface processes:** *groundwater flow, salinity and contamination, CO_2-brine interactions, time-lapse monitoring, and quantitative characterization of near-surface heterogeneity.*
- **Crustal and lithospheric dynamics:** *multi-physics imaging of crust and upper-mantle structure, melt and fluid distribution, and fluid-rock interactions that link deep Earth dynamics to resource distribution and geothermal activity.*

Peer-reviewed Publications

†Co-first author; *Corresponding author.

[22] **Wu, S.**, Sun, J., Yang, D. & Chen, J., 2026. CURIE: A continental-scale probabilistic magnetotelluric reference model of USArray data for the contiguous United States. *Geophysical Journal International*, ggag451. https://doi.org/10.1093/gji/ggag451.

[21] **Wu, S.** & Sun, J., 2026. Probabilistic inversion of electromagnetic data using Stein variational gradient descent. *Geophysical Journal International*, ggag392. https://doi.org/10.1093/gji/ggag392.

[20] **Wu, S.**, Sun, J. & Chen, J., 2025. Variational inference for geophysical Bayesian inverse problems using normalizing flows: An unsupervised approach to electromagnetic data inversion. *Geophysical Journal International*, 242(3), ggaf239. https://doi.org/10.1093/gji/ggaf239.

[19] **Wu, S.**, Thoram, S., Sun, J., Sager, W. W. & Chen, J., 2025. Characterizing marine magnetic anomalies: A machine learning approach to advancing the understanding of oceanic crust formation. *Journal of Geophysical Research: Solid Earth*, 130(2), e2024JB030682. https://doi.org/10.1029/2024JB030682.

[18] **Wu, S.**, Huang, Q. & Zhao, L., 2024. Physics-guided deep learning-based inversion for airborne electromagnetic data. *Geophysical Journal International*, 238, 1774–1789. https://doi.org/10.1093/gji/ggae244.

[17] **Wu, S.**, Huang, Q. & Zhao, L., 2023. Fast Bayesian inversion of airborne electromagnetic data based on the invertible neural network. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 5907211. https://doi.org/10.1109/TGRS.2023.3264777.

[16] **Wu, S.**, Huang, Q. & Zhao, L., 2023. A deep learning-based network for the simulation of airborne electromagnetic responses. *Geophysical Journal International*, 233, 253–263. https://doi.org/10.1093/gji/ggac463.

[15] **Wu, S.**, Huang, Q. & Zhao, L., 2022. Instantaneous inversion of airborne electromagnetic data based on deep learning. *Geophysical Research Letters*, 49(10), e2021GL097165. https://doi.org/10.1029/2021GL097165.

[14] **Wu, S.**, Huang, Q. & Zhao, L., 2021. Convolutional neural network inversion of airborne transient electromagnetic data. *Geophysical Prospecting*, 69(8–9), 1761–1772. https://doi.org/10.1111/1365-2478.13136.

[13] **Wu, S.**, Huang, Q. & Zhao, L., 2021. De-noising of transient electromagnetic data based on the long short-term memory-autoencoder. *Geophysical Journal International*, 224(1), 669–681. https://doi.org/10.1093/gji/ggaa424.

[12] Sun, J. & **Wu, S.**, 2026. Demystifying and unifying ELBO for probabilistic deep learning inversion. *The Leading Edge*, Accepted.

[11] Chen, J., Wang, L., **Wu, S.**, Sun, J. & Chen, C., 2026. Vision Transformer–based reconstruction of terrestrial water storage anomalies from GRACE/GRACE-FO over the conterminous United States. *Big Data and Earth System*, 2(3), 100075.

[10] Lu, Q.†, **Wu, S.**†, Liu, Y.†, et al., 2026. Autonomous driving chips. *Nature Reviews Electrical Engineering*, 3(9), 596–614. https://doi.org/10.1038/s44287-026-00319-1.

[09] Zhou, C., **Wu, S.***, Sun, J., Chen, J. & Huang, Y., 2026. Probabilistic inversion of airborne electromagnetic data using conditional variational autoencoder: Imaging hydrothermal systems at Yellowstone. *IEEE Transactions on Geoscience and Remote Sensing*, 64, 5908314. https://doi.org/10.1109/TGRS.2026.3681513.

[08] Huang, Q., **Wu, S.** & Xue, J., 2026. Data science and machine learning in geo-electromagnetics: A review. *Surveys in Geophysics*, 47(3), 677–732. https://doi.org/10.1007/s10712-025-09904-9.

[07] Bittar, G., **Wu, S.***, Su, Y., Zeng, S., Sun, J., Wu, X., Huang, Y. & Chen, J., 2025. Invertible neural network for real-time inversion and uncertainty quantification of ultra-deep resistivity measurements. *Computers & Geosciences*, 207, 106067. https://doi.org/10.1016/j.cageo.2025.106067.

[06] Su, Y., **Wu, S.**, Sun, J., Wu, X., Huang, Y., Chen, J., Lu, L., Wei, X. & Christiansen, R., 2025. Natural hydrogen exploration by joint sparse inversion of geophysical measurements and integrated geological interpretation. *International Journal of Hydrogen Energy*, 173, 151040. https://doi.org/10.1016/j.ijhydene.2025.151040.

[05] Xue, J., Huang, Q., **Wu, S.**, Zhao, L. & Ma, B., 2024. Real-time dual-parameter full-waveform inversion of GPR data based on robust deep learning. *Geophysical Journal International*, 238, 1755–1771. https://doi.org/10.1093/gji/ggae243.

[04] Xue, J., Huang, Q., **Wu, S.** & Zhao, L., 2024. Detection of ULF geomagnetic anomalies prior to the Tohoku-Oki Earthquake by the multireference station method. *IEEE Transactions on Geoscience and Remote Sensing*, 62, 5910009. https://doi.org/10.1109/TGRS.2024.3382472.

[03] Xue, J., **Wu, S.**, Huang, Q., Zhao, L., Sarlis, N. V. & Varotsos, P. A., 2023. RASE: A real-time automatic search engine for anomalous seismic electric signals in geoelectric data. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 5905911. https://doi.org/10.1109/TGRS.2023.3260202.

[02] Xue, J., Huang, Q., **Wu, S.** & Nagao, T., 2022. LSTM-autoencoder network for the detection of seismic electric signals. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 5917012. https://doi.org/10.1109/TGRS.2022.3183389.

[01] Wang, K., Huang, Q. & **Wu, S.**, 2020. Application of long short-term memory neural network in geoelectric field data processing. *Chinese Journal of Geophysics* (in Chinese), 63(8), 3015–3024. https://doi.org/10.6038/cjg2020O0119.

Preprints and Papers under Review or in Preparation

- **Wu, S.**, Sun, J. & Chen, J., 2026. Continental-scale probabilistic resistivity imaging of Australia using deep learning: Implications for geology, groundwater, and critical minerals. *arXiv preprint*, arXiv:2609.19516.
- Xiao, L.[†,*], **Wu, S.**[†,*], Xiao, V., et al., 2026. Structural alignment for reliable industrial AI: Bridging physical reality, data, models, and human intent. *arXiv preprint*, arXiv:2609.35400.
- Chen, J., **Wu, S.**, Wang, L., Chen, C. & Sun, J., 2025. Reconstruction of GRACE-derived TWS in the Qinghai–Tibet Plateau using convolutional encoder-decoder and multi-source data. In preparation.

Conference Papers

[23] **Wu, S.**, Sun, J. & Chen, J., 2026. Probing Australia's crust: Insights from continental-scale deep learning probabilistic inversion of AusAEM data. In *IMAGE Annual Meeting Abstracts*.

[22] Zhou, C., **Wu, S.**, Sun, J., Chen, J. & Huang, Y., 2026. Rapid probabilistic resistivity imaging of AEM data using conditional variational autoencoders. In *IMAGE Annual Meeting Abstracts*.

[21] **Wu, S.**, Sun, J. & Chen, J., 2025. Australia's shallow subsurface revealed: Fast continental-scale probabilistic inversion of AusAEM data via deep learning. In *AGU Annual Meeting Abstracts*.

[20] Chen, J., **Wu, S.**, Wang, L., Chen, C. & Sun, J., 2025. Interpretable high-resolution mapping of terrestrial water storage dynamics on the Qinghai–Tibet Plateau based on deep learning. In *AGU Annual Meeting Abstracts*.

[19] Kietzmann, J., **Wu, S.**, Sun, J. & Niu, Q., 2025. Critical zone subsurface structure characterization of a mountainous catchment using seismic refraction and drone-based electromagnetics. In *AGU Annual Meeting Abstracts*.

[18] Zhou, C., **Wu, S.**, Sun, J., Chen, J. & Huang, Y., 2025. Deep learning-based probabilistic imaging of Yellowstone's hydrothermal system from AEM Data. In *AGU Annual Meeting Abstracts*.

[17] **Wu, S.**, Sun, J. & Chen, J., 2025. An unsupervised deep learning approach to geophysical Bayesian inversion using normalizing flows. In *IMAGE Annual Meeting Abstracts*.

[16] Su, Y., **Wu, S.**, Sun, J., Wu, X., Huang, Y., Chen, J., Lu, L., Wei, X. & Christiansen, R., 2025. Geophysical targeting of serpentinite for geologic hydrogen by joint analysis of gravity and magnetic data. In *IMAGE Annual Meeting Abstracts*.

[15] **Wu, S.**, Thoram, S., Sun, J., Sager, W. W. & Chen, J., 2024. Transforming the interpretation of marine magnetic anomalies through a machine learning-based framework. In *AGU Annual Meeting Abstracts*.

[14] **Wu, S.**, Sun, J. & Chen, J., 2024. Fast model uncertainty evaluation of airborne frequency-domain electromagnetic data inversion based on deep learning. In *AGU Annual Meeting Abstracts*.

[13] Su, Y., **Wu, S.**, Chen, J., Sun, J. & Lu, L., 2024. Identifying natural hydrogen reservoirs through integrated 3D aeromagnetic and gravity data inversion in Bartlett Springs fault zone in north California. In *AGU Annual Meeting Abstracts*.

[12] Sun, J., **Wu, S.**, Chen, J. & Yin, Z., 2024. Bayesian inference of airborne electromagnetic data based on normalizing flows. In *AGU Annual Meeting Abstracts*.

[11] Huang, Q., Xue, J. & **Wu, S.**, 2024. Data science and machine learning in geo-electromagnetics. In *EM Induction Workshop Abstracts*.

[10] Kalu, D. V., **Wu, S.** & Sun, J., 2024. Empowering mineral exploration: Leveraging invertible neural networks for magnetotelluric data inversion and uncertainty quantification. In *IMAGE Annual Meeting Abstracts*.

[09] Bittar, G., Su, Y., **Wu, S.**, Sun, J., Wu, X., Huang, Y. & Chen, J., 2024. Fast inversion and uncertainty quantification of electromagnetic well logging data using invertible neural network. In *IMAGE Annual Meeting Abstracts*.

[08] **Wu, S.**, Sun, J. & Chen, J., 2024. Airborne electromagnetic data interpretation with deep learning-based stochastic inversion and posterior distribution clustering with application to salinization detection. In *GEM Workshop Abstracts*.

[07] **Wu, S.**, Huang, Q. & Zhao, L., 2023. Simultaneous resistivity imaging of airborne electromagnetic data based on deep learning. In *JpGU Meeting Abstracts*.

[06] **Wu, S.**, Huang, Q. & Zhao, L., 2023. Near real-time subsurface structure imaging using airborne electromagnetic data based on deep learning. In *EGU General Assembly Abstracts*.

[05] **Wu, S.**, Huang, Q. & Zhao, L., 2022. Near real-time resistivity imaging from airborne electromagnetic data based on deep learning. In *CGU Annual Meeting Abstracts*.

[04] **Wu, S.**, Huang, Q. & Zhao, L., 2021. 1-D inversion of airborne transient electromagnetic data based on convolutional neural network. In *CGU Annual Meeting Abstracts*.

[03] **Wu, S.**, Huang, Q. & Zhao, L., 2021. Convolutional neural network inversion of airborne transient electromagnetic data. In *CIGEW Abstracts*.

[02] **Wu, S.**, Huang, Q. & Zhao, L., 2020. De-noising of transient electromagnetic data based on the LSTM-autoencoder. In *CGU Annual Meeting Abstracts*.

[01] **Wu, S.** & Huang, Q., 2019. De-noising of transient electromagnetic data based on the LSTM-autoencoder. In *CIGEW Abstracts*.

Research Grants

[07] Grants to Enhance and Advance Research at the University of Houston. **Co-Principal Investigator**. *An integrated framework for natural hydrogen reservoir identification through geophysical, geological, and machine learning approaches*. Project period: 04/2025-10/2026. Funding: 40,000 USD.

[06] National Natural Science Foundation of China. No. 42204074. **Principal Investigator**. *De-noising of airborne transient electromagnetic data based on deep learning*. Project period: 01/2023-12/2024. Funding: 200,000 RMB.

[05] China Postdoctoral Science Foundation. No. 2022M720214. **Principal Investigator**. *Bayesian inversion of airborne electromagnetic data based on deep learning*. Project period: 01/2023-12/2024. Funding: 80,000 RMB.

[04] National Natural Science Foundation of China. No. U2239201. **Collaborator**. *Study on the spatiotemporal characteristics of seismic electromagnetic anomalies in the Sichuan-Yunnan region and their relationship with seismic activity*. Project period: 01/2023-12/2026. Funding: 2,870,000 RMB.

[03] National Natural Science Foundation of China. No. 42274088. **Collaborator**. *Study on the electrical structure of the middle to upper mantle in the Chinese Mainland based on geomagnetic diurnal variations*. Project period: 01/2023-12/2026. Funding: 560,000 RMB.

[02] National Natural Science Foundation of China. No. 41874082. **Collaborator**. *Study of joint methodology integrating wavelet analysis and probability tomography of self-potential data and the application in landslide monitoring*. Project period: 01/2019-12/2022. Funding: 650,000 RMB.

[01] National Natural Science Foundation of China. No. 41804072. **Collaborator**. *Three-dimensional forward modeling of global and regional-scale ocean tidal electromagnetic signals.* Project period: 10/2019-12/2020. Funding: 250,000 RMB.

Invited Talks

- **Continental-Scale Subsurface Resistivity Imaging of Australia: Implications for Groundwater and Critical Mineral Systems**
  MIT ERL Presents: Geo-Solutions for the Energy Transition, Cambridge, MA, 2026
- **AI-enabled electromagnetic imaging with quantified uncertainty from near-surface to deep Earth**
  Department of Geology and Geophysics, University of Wyoming, Laramie, WY, 2026
- **AI-enabled electromagnetic imaging with quantified uncertainty from near-surface to deep Earth**
  School of Earth Sciences, The Ohio State University, Columbus, OH, 2026
- **Data science for energy transition**
  Guest instructor in NSF Data Science Corps Summer Camp, online, 2025
- **Advancing subsurface characterization with deep generative models**
  Department of Petroleum Engineering Weekly Seminar, Houston, TX, 2025

- **Advancing the interpretation of potential field geophysical data with deep generative models**
  Geophysical Society of Houston Potential Fields Special Interest Group, Houston, TX, 2024
- **Data science for energy transition**
  Guest instructor in NSF Data Science Corps Summer Camp, online, 2024
- **Airborne electromagnetic data inversion and uncertainty quantification using deep learning**
  The United States Geological Survey, online, 2024
- **Generative AI for Geoscience Applications**
  Amazon Web Services, Houston, TX, 2023
- **Deep learning-based inversion and uncertainty quantification of airborne electromagnetic data**
  China Aero Geophysical Survey & Remote Sensing Center for Natural Resources, Beijing, China, 2023
- **Deep learning-based inversion and uncertainty quantification of airborne electromagnetic data**
  National Institute of Natural Hazard, Ministry of Emergency Management of the People's Republic of China, Beijing, China, 2023

Patents

[02] **Wu, S.** & Huang, Q., 2022. An inversion method of airborne electromagnetic data based on LSTM network [P]. CN113568055B, 2022-07-01.

[01] Huang, Q. & **Wu, S.**, 2022. An inversion method of airborne electromagnetic data based on convolutional neural network [P]. CN113568056B, 2022-07-01.

Honors and Awards

- GWH Award for Excellence in Electrical Geophysics *Hohmann Wannamaker Trust, 2025*
- Outstanding Doctoral Dissertation Award *Chinese Geophysical Society, 2022*
- Excellent Graduate Award *Peking University, 2021*
- Outstanding Student Presentation Award *Chinese Geosciences Union Annual Meeting, 2020*
- Hai Liang Scholarship *Peking University, 2020*
- Merit Student Award *Peking University, 2020*
- Outstanding Student Presentation Award *Chinese Geophysical Society, 2019*
- Award for Scientific Research *Peking University, 2019*
- Award for Scientific Research *Peking University, 2018*

Teaching Experience

- Graduate, **Data science for energy transition** (50 students) *University of Houston, 2025*
- Graduate, **Data science for energy transition** (50 students) *University of Houston, 2024*
- Graduate, **Electromagnetic Field Experiment** (2 students) *Peking University, 2020*
- Undergraduate, **Introduction to Seismology** (320 students) *Peking University, 2019*

Field Experiences

- **Houston, TX, USA** *2025*
  Frequency-domain electromagnetic (FEM) measurements to investigate saltwater intrusion in coastal aquifers.
- **Boise, ID, USA** *2025*
  Drone-based electromagnetic measurements to investigate the critical zone resistivity structures and groundwater systems.
- **Houston, TX, USA** *2024*
  Drone-based electromagnetic and magnetic measurements to investigate groundwater systems and processes.

- **Tangshan, China** *2020*
  Magnetic and FEM measurements, to investigate the underground structure beneath a large-scale surface collapse.
- **Jingdezhen, China** *2018*
  Magnetotellurics (MT) and controlled-source electromagnetic (CSEM) measurements, to investigate the world's largest known tungsten deposit.

Services

- **Professional Organizations and Meetings**
  Session co-chair for EM 1 New Advancements in Electromagnetic Applications, IMAGE, Houston, TX, USA, 2026
  Session co-chair for GM 2 Interpretation and Case Studies, IMAGE, Houston, TX, USA, 2025
  Session co-chair for NEF 2 Geothermal and Critical Minerals, IMAGE, Houston, TX, USA, 2025
  Session co-chair for NEF P4 Geologic Hydrogen 2, IMAGE, Houston, TX, USA, 2025
  Session convener for T001-I. Advances in machine learning applications in marine geosciences: from data acquisition and analysis to interpretation across varied marine environments and scales, AGU, Washington, D.C., USA, 2024
  Session co-convener for NS008. Advances in Multimethod Geophysical Data Interpretation and Other Applications of Machine Learning, AGU, Washington, D.C., USA, 2024
  *Guest Editor for special issue on Frontiers in Electromagnetic Geophysics, **Geophysics**, 2024*
  Session co-chair for NEF P1 Emerging Energy: Building the Future on the Past 2, IMAGE, Houston, TX, USA, 2024
  Session co-chair for EM 1 Modeling and Inversion, IMAGE, Houston, TX, USA, 2024
  Session co-chair for S-EM14 Electric, magnetic and electromagnetic survey technologies and scientific achievements, JpGU Geoscience Union Meeting, Chiba, Japan, 2023
- **Reviewer for Journals**
  Science Advances, Geophysical Journal International, IEEE Transactions on Geoscience and Remote Sensing, IEEE Geoscience and Remote Sensing Letters, IEEE Journal of Selected Topics in Signal Processing, Geophysics, Geophysical Prospecting, Earth, Planets and Space, Pure and Applied Geophysics, Chinese Journal of Geophysics.